



Real-Time Data Warehouse & Business Intelligence Project for NIDP

Project Summary

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EXECUTIVE SUMMARY

The NIDP ecosystem operates multiple independent operational systems, including Registration, Authentication, Data Verification, Grievance Management, Card Printing, Anonymous Profile, and other operational databases. Reporting has been fragmented, largely manual, and lacks a unified, trusted data foundation to support timely decision-making, operational oversight, and controlled external reporting.

To address these limitations, this project has been proposed and built with a streaming-enabled enterprise Data Warehouse platform which provides a centralized and well governed analytics environment. Operational data changes will be captured using Debezium and streamed through Apache Kafka, where they are transformed into curated datasets and integrated into a high-performance ClickHouse Data Warehouse. Audience-specific data marts will deliver standardized and analytics-ready information for internal and external stakeholders. The Metabase platform will provide secure, role-based dashboards and support controlled public transparency reporting.

This architecture establishes a single source of truth for National ID performance, enables near real-time insights, strengthens data governance, and provides a scalable foundation for evidence based decision-making.

Problem Statement

NIDP's enrolment and authentication data is growing very fast with registration surpassing 30 Million and nearly 200K daily authentications. NIDP had no centralized, scalable, and governed analytics platform to support a unified reporting/dashboard, real-time operational visibility. The existing reporting approach did not provide a single source of truth (sometimes there is duplicated data as there is no referential integrity enforced on the Dashboard database), standardized performance metrics, or controlled data access required for national-scale decision-making and transparency.

Objectives

The objective of the project is:

- To develop and implement a central data warehouse/ClickHouse that integrates real-time data from various sources and a unified, dynamic BI and dashboard to provide actionable insights and data-driven decision-making to align with the company's strategic goals. • To generate automated and dynamic reports when and where needed
- To establish a single source of truth for NID reports and dashboards.
- To deliver role-based dashboards for technical, operational, administrative, executive decision-making, partners, and public audiences.

Previous Architecture

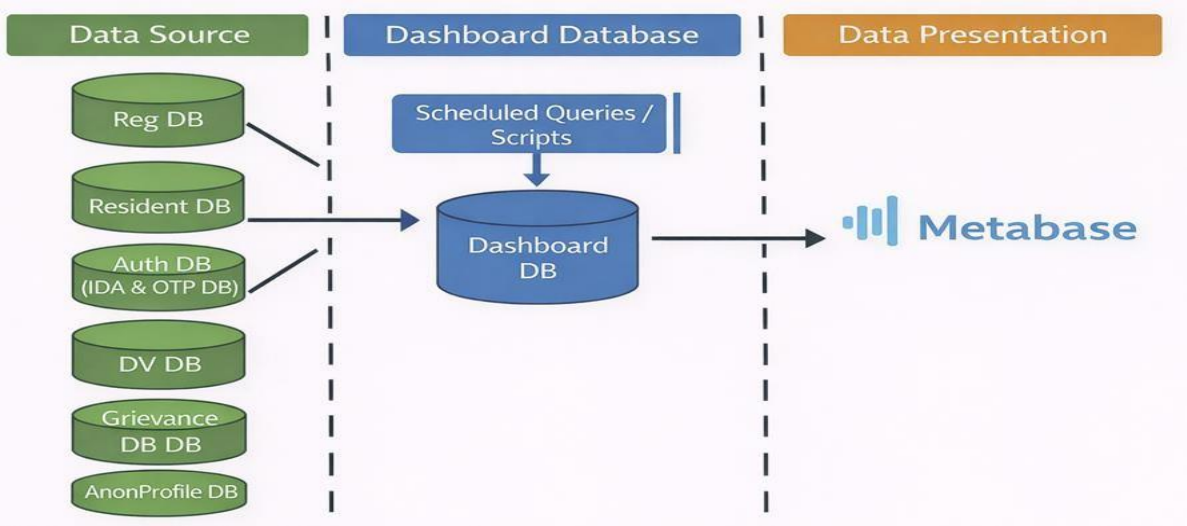


Figure 1: NIDP's old reporting architecture

The existing reporting environment relies on a standard database used as a de facto warehouse. Data is extracted from selected operational databases through scheduled queries and scripts and is visualized using Metabase dashboards. The same environment has been serving both internal teams and external partners through role-based access.

While this approach provides basic reporting capability, it was not designed to function as an enterprise analytics platform for a national-scale identity system.

Key Challenges

- Reporting logic in the Metabase doesn't consider performance. It only checks for syntax.
- Long running queries that degrade performance as the data volume is growing.
- Lack of unified and trusted enterprise data warehouse
- Same environment serves both internal users and external partners (role-based access).
    - This may create risks like:
        - ✦ Security boundary problem (configuration mistakes may occur; metadata exposure is possible ...)
        - ✦ Data governance risk (internal dashboards evolve frequently; external partners should see stable metrics only).
        - ✦ Performance & stability risk
- Limited decision visibility as all the data sources is not integrated.
- Limited Data Governance
- Even if Metabase supports role-based access, that alone is not sufficient, it can not provide partner access through a controlled portal or data service layer.

Revised Solution Architecture

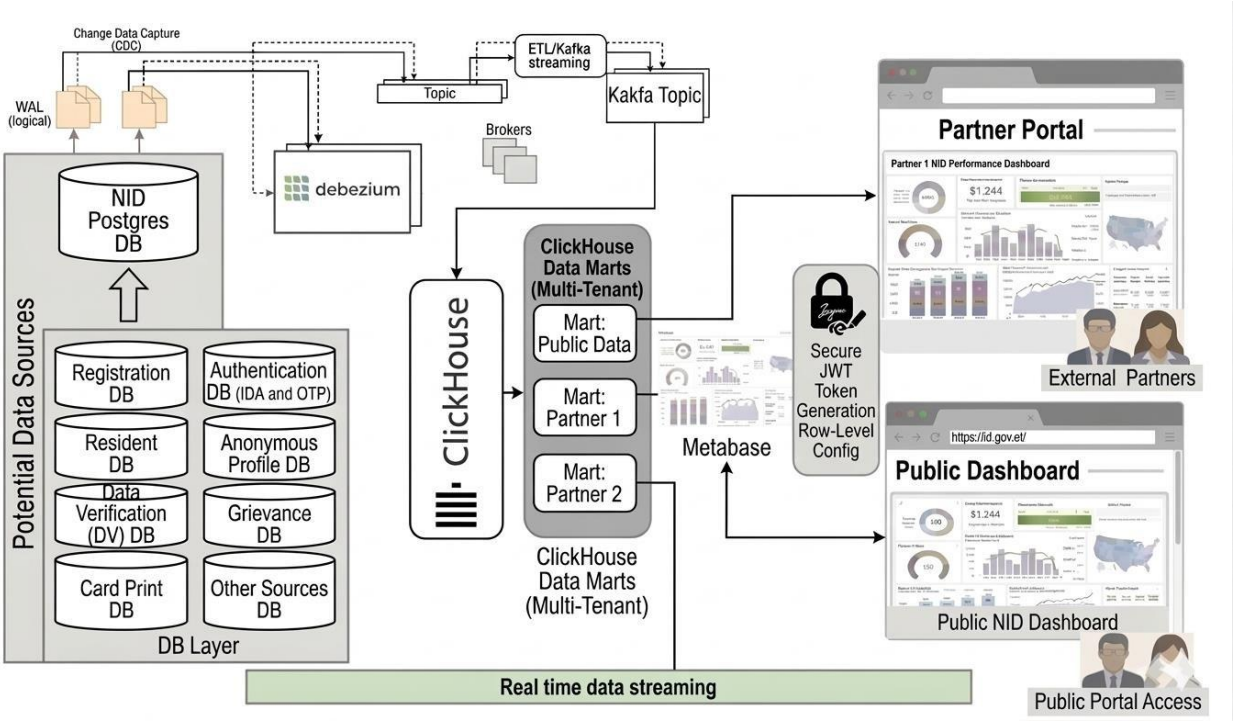


Figure 2: Revised NIDP Data warehouse and Reporting Platform Architecture

So, for security, governance, and performance reasons, external partners should not access the internal BI environment directly. Partner reporting will be delivered through a controlled portal or dedicated reporting environment that exposes only verified and aggregated data.

Flow:
Operational Databases → CDC (captures changes from WAL) → Streaming Backbone (publish events) → Curated Event Topics → ClickHouse/Data Warehouse → Data Marts (governed, audience-specific) → Metabase → Partner/Public Access Layer

Implies, operational data is captured through CDC, streamed into a centralized data warehouse, transformed into governed data marts, and delivered through role-based dashboards and controlled public reporting channels. Data marts will be implemented as materialized views within the data warehouse. These views update automatically as new data is ingested, ensuring near real-time reporting.

Core components of architecture

- **CDC:** Debezium captures database changes with minimal impact on source systems. It captures changes directly from database transaction logs such as PostgreSQL WAL. Connectors read these logs and emit structured change events—insert, update, delete—to

Kafka topics. This approach avoids polling, minimizes load on source databases, and preserves event ordering.

- **Streaming Backbone:** Apache Kafka provides reliable, scalable data transport and replay. Streaming layer transport and distributions events; transformations occur during ingestion and warehouse processing.
- **Enterprise Data Warehouse:** ClickHouse stores integrated, historical, analytics-ready data (analytics storage).
- **Presentation Layer:** Metabase delivers secure, audience-specific dashboards.

Data marts provide governed, performance-optimized views of warehouse data tailored to specific audiences. They simplify access control, improve dashboard performance, and prevent direct querying of raw warehouse tables.

Key Benefits

- Single trusted source of Analytics and BI for national ID and key stakeholders
- Improved data-driven decision making (leadership gains reliable analytics for strategic planning)
- Near-real-time operational visibility
- Standardized and automated reporting
- Scalable foundation for advanced analytics
- Secure role-based data access

NB. This architecture is adapted from MOSIP and used for **near-real-time reporting** and is modern, scalable, and aligns with what large digital ID ecosystems use. It is a **real-time Data Warehouse architecture** rather than traditional batch ETL

Data Sources for the BI and dashboard

1. Registration Data
2. Resident Services Data
3. Authentication Data (IDA and OTP DBs)
4. Anonymous Profile Data
5. Data Verification (DV) Data
6. Grievance Management Data
7. Card Print Tracking Data
8. Other relevant data sources

Key Metrics that highlight critical performance indicators (KPIs)

Selecting the right metrics for BI and dashboards is vital for providing actionable insights and supporting strategic decision-making. We need to determine which metrics are important for the NIDP's dashboard and dynamic reports. We can consider the following key metrics:

1. Enrollment & registration performance

    - Total registrations (daily / monthly / cumulative)
    - New registrations per day
    - Registrations by Region
    - Registration success and rejection rate (%)
    - Duplicate detection rate (%)
    - Biometric capture failure rate (%)
    - Other key metrics

2. Authentication Metrics

    - Total Authentication Requests
    - Auth Requests per Day/hour/
    - OTP Requests per Day
    - Authentication success and failure rate (%)
    - Other key metrics

3. Data Verification (DV) performance

    - Total Verification Requests
    - Verification approval rate (%)
    - Verification rejection rate (%)
    - Other key metrics

4. Card Production and delivery
    - Total number of cards printed by partner
    - Cards printed by age range
    - Cards printed by location
    - Number of card Re-print orders
    - Number of Backlog print requests
5. Grievance & support performance
    a. Total grievances submitted
    b. Grievance by category
    c. Resolution, escalation and reopen rate (%)
    d. Othe key metrics
6. Other metrics

Note: What KPIs should be integrated with the BI and dashboard) are not exhaustive lists, and it requires the collaboration of all the stakeholders.

Project Plan for the data warehouse, BI, and dashboard

Implementation Approach: Given the various data sources (mentioned above) to be integrated to central data warehouse (ClickHouse) to produce a unified, dynamic, and comprehensive BI and dashboard using a phased implementation. Approval is requested to initiate design and implementation. (This is just tentative duration.)

POC Environment: A testing environment will be established to validate the end-to-end data pipeline prior to production deployment. The environment will simulate operational data ingestion, streaming integration, warehouse processing, and dashboard delivery.

Resources Allocation

Overall, developing a centralized data warehouse and a unified, dynamic BI and dashboard, especially one that integrates with multi-data sources, and displays real-time data and generates dynamic reports, is a huge project by itself and requires a combination of skills and expertise. Accordingly, the following resources are required.

Core data platform team

- One DBA (Kafka, CDC, pipelines, warehouse modeling, marts, ClickHouse performance & storage)
- One DevOps Engineer (deployment, monitoring, scaling)

Dashboard & presentation team

- One BI Developer (Metabase dashboards & reports)
- One UI/UX Designer (public dashboard only)

Assigned Team roles and responsibilities

1. BI developer: Create dashboards (both internal and partner specific dashboards), reports, and semantic data models
2. DevOps engineer: Deploy infrastructure, manage CI/CD, monitoring, scaling.
3. UI/UX designer: Design public dashboard experience, integrate/embed public dashboard to our website, interactive design, Add basic branding and coloring, user journey...etc...
4. Database engineer: Optimize ClickHouse storage, indexing, and query performance, Design CDC pipelines, Kafka topics, star schemas, data marts, and ingestion workflows.

Execution Phases:

Phase 1 — Foundation & architecture

Activities

- Requirement refining and KPI definition
- Data source & existing dashboards analysis
- Gap analysis (current vs target architecture)
- Infrastructure provisioning (Kafka, ClickHouse, BI platform)
- Data governance framework
- ClickHouse setup/ deploy warehouse cluster
- Deploy Apache Kafka cluster
- Configure Debezium connectors (initial CDC setup)

Deliverables

- KPI catalog
- Data governance model
- Running base platform (Kafka + ClickHouse ready)

Phase 2 — Data ingestion & warehouse build

Activities

- Initial historical data backfill (large datasets)
- Implement Change Data Capture (Debezium)
- Configure streaming data pipelines (Kafka)
- Integrate operational systems
- Data validation and reconciliation
- Build centralized Enterprise Data Warehouse (Clikchouse warehouse)
- Data marts creation (internal, partner, public)

Deliverables

- Operational Data Warehouse
- Validated and stable CDC pipelines
- Documented schema

Phase 3 — BI & Dashboards Integration

Activities

- Metabase integration with ClickHouse
- Develop role-based dashboards
- KPI visualizations and validation
- Performance optimization
- Access control configuration
- Validate metrics with stakeholders

Deliverables

- Internal dashboards
- Partners dashboards

Phase 4 — Public dashboard /Embedded iFrame (Metabase Signed)

Activities

- Requirements refinement with stakeholders
- Public data model creation (aggregated, anonymized)
- UI/UX design (wireframing and prototyping)
- Metabase embedding (signed embed)
- Implement authentication and security controls

Deliverables

- UI/UX design
- Public dashboard portal

Phase 5 — Final Testing and deployment

Activities

- End-to-end integration testing
- Performance testing
- Data accuracy validation (source vs warehouse)
- Security testing
- Production deployment
- Monitoring setup (Prometheus + Grafana)

Deliverables

- Production platform
- Monitoring dashboards • Operational documentation.

Potential challenges/constraints with the new architecture

- Infrastructure provisioning (requires more infrastructure) as the data in the DW grows rapidly
- Large data volume (especially IDA ~20TB)
- Historical data backfill (CDC tools capture new database changes)
- Integration complexity as there are multiple operational systems
- Availability of human resources